

Why Public-Safety Wages May Rise Faster Than Other Municipal Wages

Gabriel Wages | PI Report | July 30, 2026

18,554 valid rated spans	11,548 normalized quantitative records	416 quantitative growth records	4 bounded local candidates
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Claim boundary. Bounded local documentary evidence, not final wage-gap estimates; no national prevalence or causal claims.

1. Executive Summary

The evidence now supports a strong, directional account of municipal wage formation. Within the processed rated corpus, the records lean toward safety-side upward wage pressure in several important channels—notably non-base compensation, the volume of automatic-growth evidence, and direct market/staffing wage-pressure records. That is the direction of the documentary evidence after side classification; it is not a statement about national prevalence. The same corpus also contains meaningful non-safety counterevidence, and exact-role validation shows that a nominal safety advantage can disappear or reverse when schedule position is aligned. The central empirical implication is therefore specific: the strongest current case for a safety/non-safety disparity lies in the accumulation of occupation-specific pay-setting mechanisms, while the clean wage-level comparison base remains narrow.

Non-base compensation is the clearest safety-heavy mechanism layer. Of 3,718 valid non-base signals, 1,239 are resolved as police, 280 as fire, nine as combined safety, 226 as non-safety, and 1,964 remain unclear. Among the 1,754 records whose side is resolved, safety accounts for 87.1 percent. The evidence does include real non-safety examples—such as a Port Huron Township clerical health-opt-out stipend increasing from \$175 to \$400 per pay period—so non-base pay is not uniquely a safety mechanism. Yet the resolved layer contains far more safety-side evidence, including a 5 percent police shift differential in Old Tappan and firefighter longevity pay of \$1 per hour for every five years in Miami, Ohio. If those richer or more numerous components persist after unit-cycle matching, they would put upward pressure on safety total compensation and widen a base-plus-premium gap. The present evidence establishes that hypothesis much more sharply than a base-wage-only design, while stopping short of a matched total-compensation estimate.

Normalization and matching have already been run, and they support current claims—not just future potential. The direct-value layer includes 2,687 wage/value signals, but only 661 are explicitly schedule-tagged and just six schedule-tagged spans are fully normalized. At the candidate level, one comparison is PI-usable and three are conditional. In Shreve, Ohio, the same 2024 ordinance lists part-time police officers at \$22 per hour and a part-time utility clerk at \$16 per hour: a bounded, source-grounded safety-side difference of \$6 per hour, or 37.5 percent relative to the clerk rate. Canastota points the other way: its Police Officer Year 1 rate is \$0.91 per hour below the code-enforcement rate. These records show both that local listed-rate differences exist and that rank, step, and comparator choice can change their direction.

The growth-mechanism audit materially changes the interpretation of the strongest examples. The rescue layer had identified 416 quantitative growth-mechanism records, but report-use validation shows that some prominent examples were misclassified. Phippsburg's 3.2 percent item belongs to an assessing agent, not police; Tonka Bay's cited schedule is for maintenance employees; and Howland fire percentages referring to employee insurance cost-sharing are not wage increases. The final report uses only context-validated raises, COLA/index provisions, step rules, or payment-timing mechanisms. The resulting evidence remains safety-heavy, but it contains fire and non-safety examples and therefore permits actual comparison of mechanism form rather than a police-only list.

Bargaining and market-pressure evidence clarifies how a disparity could be produced. Collective bargaining can set base schedules, premiums, and effective dates; interest arbitration and factfinding can shape prospective terms during impasse; settlement converts proposals into amounts and timing; and no-strike rules alter leverage. These pathways are not interchangeable with grievance arbitration, which generally enforces an existing agreement. The bargaining audit is safety-heavy in volume—714 safety versus 292 non-safety spans—but only 77 of 1,716 audited spans directly link the institution to wages, so the sign of the average effect remains unresolved. Market/staffing evidence also leans safety among resolved direct records (33 safety, four non-safety, 46 unclear), while Brewster dispatchers provide a strong non-safety counterexample: a source-reported 19.4 percent base-pay increase over three years expressly justified by recruitment, retention, and operational stability.

The report's central conclusion is bounded but substantive. The current record favors a safety-side interpretation for the composition and documentary concentration of several pay-growth channels, while matched wage-level evidence is mixed and thin. It supports saying that safety compensation can be pushed upward through multiple additive channels and that these channels are plausible sources of widening. It also supports saying that non-safety market responses and schedule locations can narrow or reverse particular contrasts. The immediate empirical priority is review of the small comparison-ready stratum, followed by occupation-, schedule-location-, and cycle-aligned expansion. Final wage-gap estimation, population-level statements, and causal attribution remain outside what this corpus currently establishes.

2. Processed Evidence Base

The report draws on a broad documentary pipeline but treats the cross-occupation matched unit-cycle as the analytic target. Scout coverage reaches 16,887 municipalities, or 47.45 percent of the project's 35,589 eligible/known municipality universe. In the 4×2,500 wave, 10,000 terminal outcomes produced 9,968 parseable results, 9,977 raw candidate rows, 9,072 deduplicated candidates, 5,768 verification-ready records, 3,950 source-review-ready locators, and 3,672 retained sources. Text extraction produced 2,795 clean span-ready sources; span extraction produced 19,118 candidates, of which 18,612 entered rating. The valid rating ledger contains 18,554 spans and excludes 58 quarantined outputs.

The quantitative layer contains 11,548 normalized records: 672 full, 1,563 partial, 3,769 mechanism-only, 720 deferred for manual review, and 4,824 unusable for structured wage-level comparison. Rescue added 11 full normalizations, identified 185 near-gap-ready records, and produced 416 quantitative growth-mechanism records. Matching generated 2,712 municipality-cycle groups, 65 safety/non-safety cycle candidates, 303 structurally comparable normalized wage candidates, and 141 cycle-to-cycle growth-readiness candidates. Exact-role review then reduced current PI-facing wage comparisons to one usable and three conditional candidates. The reduction is not a failure of normalization; it is the consequence of applying the design's actual observation unit-one occupation unit, one cycle, one city-plus schedule-location comparability.

3. Codified Evidence Categories

The valid rated evidence contains 472 core-finding-ready spans, 528 supporting examples, 6,860 context-only spans, 5,533 records routed to downstream normalization, and 5,161 exclusions. Claim relevance divides into 6,454 direct quantitative supports, 6,509 mechanism-summary supports, 430 directional hints, and 5,161 weak or unsupported records.

The comparison audit does not treat those counts as independent municipalities or events. It reclassifies side, pressure direction, and exact mechanism form for the major findings:

Mechanism	Side conclusion	Pressure direction	What the evidence shows now
Non-base compensation	Safety-heavy among resolved records; non-safety examples confirmed	Upward safety total-compensation pressure; potentially gap-widening	1,528 resolved safety spans versus 226 non-safety spans; 1,964 additional spans remain side-unclear.
Base schedules and listed rates	Mixed	Observed safety advantage in Shreve/Cammack; non-safety advantage in Canastota	2,687 direct value signals; 661 explicit schedule tags; six schedule-tagged spans fully normalized; one validated and three conditional local comparisons.
Implementation, retroactivity, and payment form	Safety-heavy documentary layer; comparative sign not established	Timing/realization effect; recurring increases compound, lump sums do not	The audit distinguishes recurring base increases, retroactive base payments, one-time lump sums, delayed implementation, and unresolved payment form.

Mechanism	Side conclusion	Pressure direction	What the evidence shows now
Automatic raises, COLA/CPI, and steps	Safety-heavy audited claim layer, with fire and non-safety counterexamples	Upward pressure on the covered side; gap direction depends on the paired unit's rule	The rescue produced 416 quantitative mechanism records, but the report-use audit removes misclassified insurance, occupation, and proposal fragments before comparison.
Bargaining and dispute resolution	Safety-heavy in volume; direct wage-linked evidence remains mixed/unclear	Potential upward or constraining pressure depending demands, offers, award, and settlement	1,716 audited spans: 77 direct wage-linked, 1,578 context, 61 false positives. Safety 714, non-safety 292, mixed 39, unclear 671.
Rank, step, specialization, and classification	Mixed	Can widen or narrow listed-rate differences depending schedule location	Canastota's Year 1 police rate is below its code-enforcement rate, while higher police steps could reverse the contrast; Alburdis shows how command rank can inflate a nominal gap.
Market, recruitment, retention, and staffing	Safety-heavy among resolved direct spans, but present on both sides	Upward pressure on the occupation targeted by a market adjustment or retention response	Among 83 direct wage-pressure spans, 33 are safety, four non-safety, and 46 unclear. Brewster dispatch is a strong non-safety example.
Bounded local listed-rate differentials	Mixed	Safety advantage in Shreve/Cammack/Alburdis; non-safety advantage in Canastota	Shreve is PI-usable; Cammack and Canastota are conditional; Alburdis is limits-only.

Average and median strength values remain available in the prior technical appendix, but this report gives priority to exact-side validation and direct mechanism linkage because strength scores cannot repair a false occupation classification. That choice removed several superficially strong but substantively irrelevant spans.

4. Findings

4.1 Non-base compensation: the resolved evidence leans safety and therefore matters for the gap

Side and pressure direction. The resolved evidence is safety-heavy, and the implied pressure is upward on safety total compensation. Police, fire, and combined-safety records account for 1,528 of 1,754 side-resolved non-base signals (87.1 percent), compared with 226 non-safety signals. Another 1,964 signals remain side-unclear, so that share describes the resolved evidence layer rather than municipalities generally.

This is not a safety-only channel. The audit found non-safety longevity, shift, overtime, allowance, certification, and stipend provisions across 100 sources and 96 municipality-state combinations. Port Huron Township's clerical settlement increased the health-coverage opt-out stipend from \$175 to \$400 per pay period. The counterevidence matters: a study that treats every premium as safety-specific would be wrong.

The stronger inference is compositional. Old Tappan provides a police shift differential equal to 5 percent of gross annual salary; Miami, Ohio, documents firefighter longevity pay of \$1 per hour for every five years of service; South Zanesville records a \$350 annual allowance for part-time police and an assistant fire chief. These mechanisms can add to compensation without appearing in a single base-rate comparison. If aligned city-cycle records confirm more or richer safety-side components, the effect would be gap-widening in total compensation even when base schedules move similarly. The current data show the channel and a safety-heavy documentary concentration; they do not yet sum all components into a matched total-compensation gap.

4.2 Normalized schedules already support bounded comparisons

Side and pressure direction. The observed local direction is mixed: Shreve and Cammack favor safety, Canastota favors non-safety at the selected schedule location, and Alburtis is not a valid bargaining-unit contrast. A future-only statement that comparison “requires normalization” is now obsolete. Normalization has been completed for the quantitative layer and rescue has been run. What remains limited is the number of records passing occupation, schedule-location, employment-status, and operative-period gates.

The valid rated layer contains 2,687 direct base-wage/value signals, not 2,687 complete schedules. Of these, 661 carry an explicit schedule tag and only six schedule-tagged spans are fully normalized. The distinction matters because a heading or one schedule fragment cannot identify both sides of a comparison. Still, the candidate-level validation supports a present result: Shreve's 2024 ordinance lists part-time police at \$22 per hour and a part-time utility clerk at \$16 per hour, a \$6 difference on the current hourly basis. This is direct outcome-side evidence that the ordinance assigns a higher listed rate to the police position.

Canastota is the necessary counterexample. Its Police Officer Year 1 rate of \$23.91 per hour is below the \$24.82 code-enforcement rate. That does not establish a general non-safety advantage; it shows why schedule location is part of the estimand. Comparing an entry police step with an unlabeled or single non-safety rate can understate or overstate the difference. A responsible comparison therefore pairs entry with entry, top with top, or the same tenure/grade concept and reports the selected location explicitly.

4.3 Implementation, retroactivity, and payment form change realized wage paths

Side and pressure direction. The documentary timing layer is safety-heavy, but the comparative sign remains unclear because few exact-role, same-cycle pairs contain timing terms for both sides. The economic direction of each payment form is nevertheless identifiable. A recurring base increase raises current pay and the base for later percentage increases; a retroactive base payment restores compensation for an earlier effective date; a one-time lump sum raises current cash without permanently lifting the schedule; and delayed implementation postpones realization.

The analysis treats these forms separately. A statement that an agreement is “retroactive” is not enough: the record must identify the covered unit, effective period, and whether the payment enters base wages. This is where matched structuring was already useful. It links timing evidence to municipality-cycle candidates and exposes the remaining blocker: most candidates do not contain a verified payment-form field on both the safety and non-safety sides. The report can therefore state how a mechanism pushes wages and which side receives it in a source record, but only a small subset permits a comparative timing claim.

4.4 Automatic raises and indexed rules create growth; the earlier police-only examples overstated the audit

Side and pressure direction. The audited report-use layer is safety-heavy, so the documentary evidence leans toward upward safety wage pressure. But growth mechanisms exist for fire and non-safety units as well, and matched examples can be mixed. A scheduled percentage, COLA/index rule, or step increment mechanically raises the covered schedule; the gap widens only when the safety-side rule is larger or operates more often than its same-cycle comparator.

The rescue identified 416 quantitative mechanism records-336 percentage, 49 COLA/CPI, 29 step-schedule, and two retroactive/lump-sum-but the comparison audit corrected the examples before reuse. Phippsburg's 3.2 percent item is for an assessing agent, Tonka Bay's cited schedule is for maintenance employees, and Howland fire percentages describing employee insurance contributions are not wage raises. Bath's 2.25-to-2 percent language is an offer/pattern fragment coupled with first-step elimination and cannot be presented as a simple enacted step increase.

The corrected evidence supplies genuine context. Coleraine's police agreement states a 5 percent increase effective January 1, 2024. Sunrise Beach describes a 1.5 percent police step percentage as a COLA. Marion's firefighter agreement reports general wage increases of 1.75, 1.85, 2.75, and 3.25 percent across specified agreement years. Bass Lake's minutes describe a 31 percent clerk increase to adjust to the area's going rate, a large non-safety market-correction example that requires meeting/action context. Howland contains an overlapping police 2 percent wage provision and Road Department 3 percent hourly-wage provision; on those source-reported mechanisms, the non-safety increase is one percentage point larger and exerts bounded gap-narrowing pressure for that interval. The fire percentages originally selected are insurance cost-sharing, so Howland is not a clean three-unit wage-growth comparison. These examples show upward pressure on the covered unit and also show why exact context-not a bare percentage-is decisive.

4.5 Bargaining and dispute resolution matter because they allocate value, timing, and leverage

Side and pressure direction. The institutional record is safety-heavy in volume but mixed in wage direction. Of 1,716 audited bargaining/dispute spans, 714 are safety, 292 non-safety, 39 mixed, and 671 unclear. Only 77 directly link an institution to a wage term; 1,578 are context and 61 are false positives. The data therefore favor saying that safety units are prominent in the observed institutional evidence, not that bargaining systematically produces a larger safety raise.

The mechanism is concrete. Collective bargaining determines base schedules, premiums, effective dates, and tradeoffs. Interest arbitration can determine or recommend prospective compensation when bargaining reaches impasse; factfinding can introduce comparability analysis and settlement recommendations. Settlement turns demands and offers into operative values, often with retroactivity. Grievance arbitration is different: it ordinarily interprets or enforces an existing agreement and should not be coded as a new wage-setting event without direct language. Strike/no-strike provisions alter leverage but have no fixed wage sign on their own.

Source spans illustrate both direction and constraint. Adams, Minnesota, records a clerk-negotiated 2.5 percent increase for city workers. Safety-side interest-arbitration records include specific union demands and employer offers, showing how prospective rates enter adjudication. A demand is upward pressure from the union; an employer offer can constrain it; the award or settlement determines the realized term. That sequence is the gap-relevant pathway. What remains missing for a comparative effect is the matched non-safety bargaining outcome in the same city-cycle.

4.6 Rank, step, specialization, and classification can generate or erase apparent differences

Side and pressure direction. The pressure is mixed. A steeper safety ladder, specialty premium, or command-rank schedule can widen a difference; a higher non-safety grade or a police entry step can narrow or reverse it. The relevant comparison is not "police versus any municipal rate." It is a declared schedule location on each side.

A police Year 1 step is not automatically comparable to a non-safety single rate because the latter may represent an actual rate, a maximum, an experienced incumbent, or an entirely different progression rule. Canastota demonstrates the problem rather than defeating comparison: its entry police value is below code enforcement, so the correct next comparison is entry-to-entry or a full schedule profile. Alburtis shows the other distortion: a Chief of Police outside the bargaining unit is a command-position rate, not a police-unit observation.

The operational method is now explicit. A pair must share municipality and operative period; pay basis; base/non-base concept; entry/top/step/grade location; full-/part-time status; occupation and role family;

rank/tenure; legal and bargaining-unit scope; and source validation. A failure does not erase the source-it changes the use from an analytic pair to a named-position contrast or mechanism context.

4.7 Market and staffing pressure leans safety in resolved direct evidence, but non-safety corrections can be large

Side and pressure direction. Among the 83 direct market/staffing wage-pressure spans, 33 are safety, four non-safety, and 46 unclear. The resolved direct evidence therefore leans safety, which is consistent with gap-widening pressure where police/fire recruitment or retention adjustments exceed the paired non-safety response. It is not exclusively safety-side. Brewster's non-sworn dispatch settlement reports 19.4 percent base-pay increases over three years expressly linked to recruitment, retention, professional demands, and operational stability. Bass Lake's clerk increase and Mantorville's \$5-per-hour engineering adjustment similarly show upward non-safety pressure.

Market mechanisms operate through identifiable actions: base-pay adjustment, market percentage, recruitment/hiring bonus, retention bonus, or schedule realignment against comparable jurisdictions. Sherburn records police wage-plus-market adjustments; Lincoln records a 1.5 percent office-staff retention increase; Gilbert records a 3 percent retention bonus. These are wage-pressure pathways, not merely rhetoric.

Exact-span review also improved reliability. Bernards "records retention" concerned a job title and public records; Greenwich "retention" concerned stormwater basins. Both are excluded. After that repair, the correct conclusion is stronger and narrower: genuine market pressure can raise either side, while the resolved direct evidence currently contains more safety cases.

4.8 Bounded wage differentials show outcomes; mechanisms explain how those outcomes may have formed

Side and pressure direction. The four validated candidates are mixed: three listed contrasts favor safety and one favors non-safety, but only Shreve is presently PI-usable. They show that occupation-specific rate setting produces local differentials; they do not identify a single causal channel.

Place	Period	Safety record	Non-safety record	Difference	Status	Mechanism reading	Key caveat
Shreve, OH	2024	\$22.00/hour - part-time police officer	\$16.00/hour - part-time utility clerk	+\$6.00/hour (+37.5%)	PI-usable supporting example	Same ordinance assigns a higher base rate to the police position.	Part-time status aligns, but duties, experience, schedule, and qualifications do not.
Cammack Village, AR	2024	\$25.00/hour maximum - part-time patrolman	\$20.00/hour maximum - administrative assistant	+\$5.00/hour (+25.0%)	Conditional/manual review	Authorized pay ceilings favor the patrol position.	Maximum authorized-not confirmed actual-rates; enactment fields require confirmation.
Canastota, NY	2023-2024	\$23.91/hour - Police Officer Year 1	\$24.82/hour - Code Enforcement Officer	-\$0.91/hour (-3.67%)	Conditional counterexample	The selected entry location does not favor police and shows schedule-location sensitivity.	Year 1 police step versus a single rate with no matched tenure label.
Alburtis, PA	Jan.-Jun. 2018	\$33.57/hour - Chief of Police	\$11.22/hour - Administrative Assistant	+\$22.35/hour (+199.2%)	Limits/appendix only	Command-rank and ordinance rate-setting, not a bargaining-unit comparison.	Chief was outside the police bargaining unit; role, rank, hours, and schedule location do not match.

Shreve is outcome-side evidence of ordinance rate differentiation. The ordinance itself does not tell us whether bargaining strength, recruitment pressure, job requirements, political priority, or another channel produced the \$6 difference. It becomes mechanism-relevant when paired with the broader evidence

showing how schedules, premiums, market adjustments, and bargaining terms are created. Cammack points to an authorized safety-side pay ceiling but not actual pay. Canastota shows a non-safety-favoring contrast at the selected schedule locations and therefore prevents a one-direction narrative. Alburtis illustrates why command rank and bargaining-unit status can manufacture an apparently enormous differential and is retained only as a limits example.

4.9 Matching has been optimized into comparison-readiness tiers

Side and pressure direction. Matching is a measurement rule rather than a wage-pressure mechanism. It determines whether a directional contrast is credible. The 303 upstream candidate pairs included 27 four-flag “ready” pairs, but exact-span review found false occupation classifications among that structurally compatible subset. The correct promotion rule is exact-role validation, not flag accumulation.

T i e r	Comparison state	C o u n t	Unit	Current use	Main blocker
1	Position/schedule-location comparable	1	validated candidate	Bounded local statement after source and calculation review	No final estimator or role-equivalence claim
2	Conditional named-position comparison	3	validated candidates	Supporting/counterexample/limits with explicit candidate caveat	Maximum/entry/rank/bargaining-status mismatch
3	Structurally compatible pairs	27	candidate pairs	Exact-role validation queue	Structural flags include false occupation classifications
4	Manual-review pairs	87	candidate pairs	Targeted position/step/period repair	One or more comparability gates unresolved
5	Not comparison-ready	189	candidate pairs	Mechanism context only or exclude	Low normalization quality or incompatible fields

The immediate analytic stratum is thus small but real. Tier 1 supports a bounded local statement. Tier 2 supplies qualified examples and counterexamples. Tier 3 is the highest-value manual review queue because pay basis and periods are largely aligned but occupation, schedule location, or legal status still need direct confirmation. The 185 near-gap-ready records are record-level inputs rather than 185 comparisons; they must be paired under the same gates.

5. Limits

The report takes the strongest direction supported by the current evidence, but it does not relabel documentary selection as population evidence. Safety-heavy span counts may reflect source availability, document type, extraction, or the greater textual complexity of safety compensation. They identify where the processed evidence pulls; they do not establish how common a mechanism is across municipalities.

No final wage-gap estimate is reported. Shreve is a source-grounded local listed-rate comparison; Cammack and Canastota remain conditional; Alburtis is limits-only. The project has not estimated a cross-city mean, weighted municipal effect, total-compensation gap, or nationally representative difference. The normalized layer still has unresolved rank, step, work-status, and cycle fields, and no analyst-side cost-of-living adjustment was performed.

No final causal conclusion is made. Bargaining, interest arbitration, market adjustment, non-base compensation, and timing have clear wage-pressure pathways, but the current records do not isolate counterfactual effects. No regression or treatment-effect model was run. COLA/CPI clauses are treated as source-reported contractual growth mechanisms, not as analyst-side cost-of-living adjustments.

The report excludes 601 OCR-later sources and does not infer content from them. Search, reachability, retention, extraction, rating, and normalization filters shape the corpus. Exact-span auditing also revealed

false positives and occupation misclassifications in the prior report-use layer; the final audit removes or corrects them rather than treating the original rating as infallible.

6. Current Scout Wave Status

The 4×2,500 wave has been processed through scouting, candidate review, verification, source review, readiness, text extraction, span extraction, rating, ingestion/codification, normalization, matching, rescue, bounded comparison validation, and the comparison/mechanism audit. The public dashboard remains organized around scout coverage rate-16,887 of 35,589 eligible/known municipalities, or 47.45 percent-with the raw count shown only as context. Technical details remain collapsed. Public dashboard: <https://dkyaya.github.io/gabriel-wages/>

7. Recommended Next Steps

1. **Review the send-ready report with the PI.** Confirm the side-direction statements, the removed false positives, and the corrected growth examples before external circulation.
2. **Manually validate Tier 2 and Tier 3 comparisons.** Confirm enacted status, occupation, schedule location, rank, work status, and bargaining scope before promotion.
3. **Build paired schedule profiles.** For the 65 matched cycles and highest-value 303 candidate pairs, represent entry, midpoint, and top rates for both safety and non-safety units rather than selecting one arbitrary rate.
4. **Add non-base symmetry.** Align longevity, shift, hazard, certification, overtime-related, stipend, and allowance components by city-cycle so the safety-heavy mechanism finding can be tested as a total-compensation difference.
5. **Separate payment form in the timing layer.** Code recurring base, retroactive base, one-time lump sum, and delayed implementation as distinct comparison fields.
6. **Address the 601 OCR-later sources in a separately authorized phase.** Do not mix that deferred intake with the current validated evidence.
7. **Delay causal modeling.** A final wage-gap estimator and any identification strategy should follow-rather than precede-the validated matched structure.